NEOSTATS RISK

AI-Powered Credit Risk Intelligence Platform

Home Credit Default Risk · AI Engineer Assignment · NeoStats

Intelligence. Innovation. Impact.

ML Risk Scoring

SHAP Explainability

NL-to-SQL Chatbot

Docker Deploy

The Problem

Banks face growing pressure to make credit decisions that are fast, accurate, and explainable.

Too Slow

Manual reviews take days — customers expect instant decisions.

High Default Rates

Poor risk scoring leads to bad loans and financial loss.

No Explainability

Black-box models fail auditors and regulators.

Analysts Can't Self-Serve

Every data question goes through IT, creating bottlenecks.

Our Solution

One end-to-end AI platform — from raw data to explainable, auditable credit decisions.

EDA

Deep portfolio insights

Talk-to-Data

Ask in plain English

ML Model

Predict default risk

Explainable AI

SHAP-powered reasons

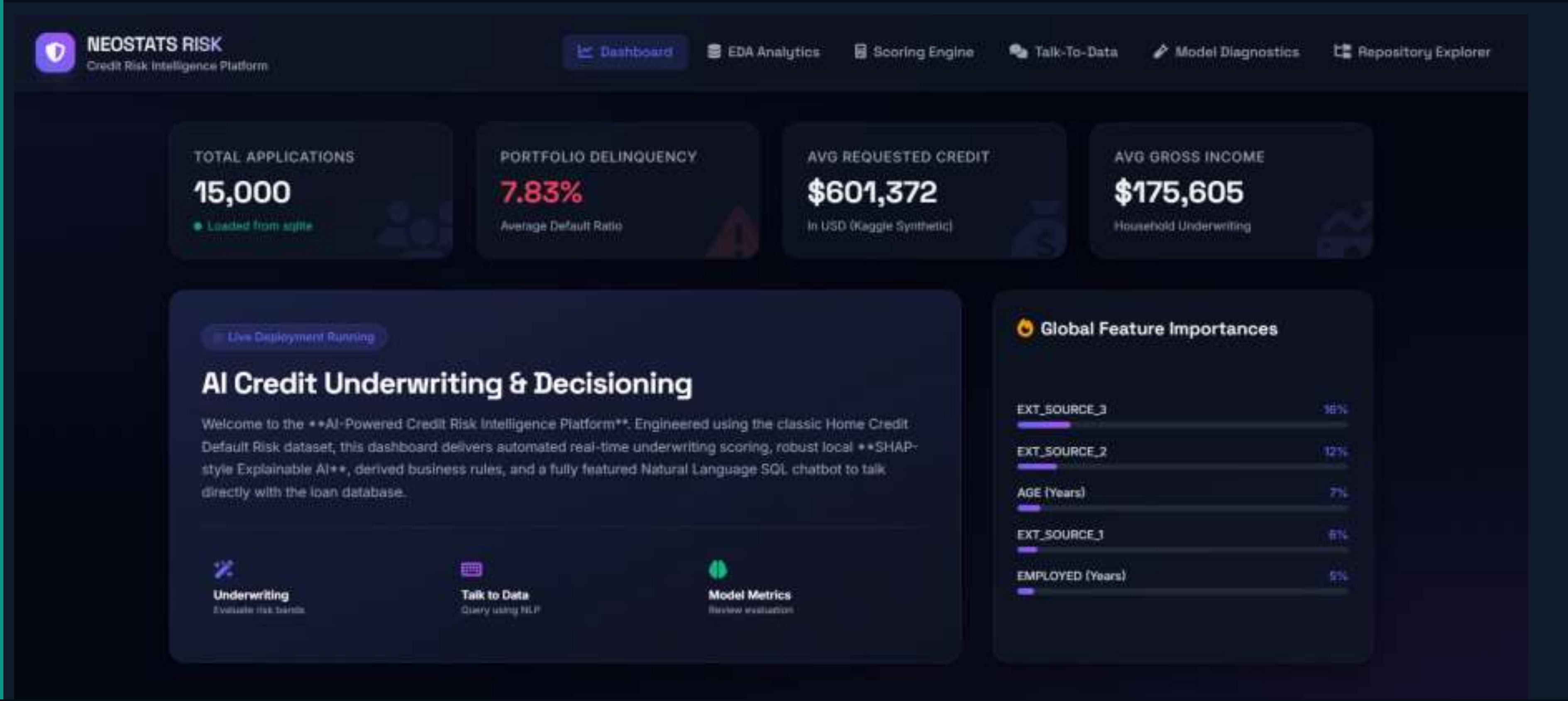
Decision Rules

Human-readable policy

Clean UI · Dockerized · Runs with a single command

Dashboard — Portfolio Overview

Live App Screen



KPIs: 15,000 applications · 7.83% default rate · Global SHAP feature importance · Live deployment status

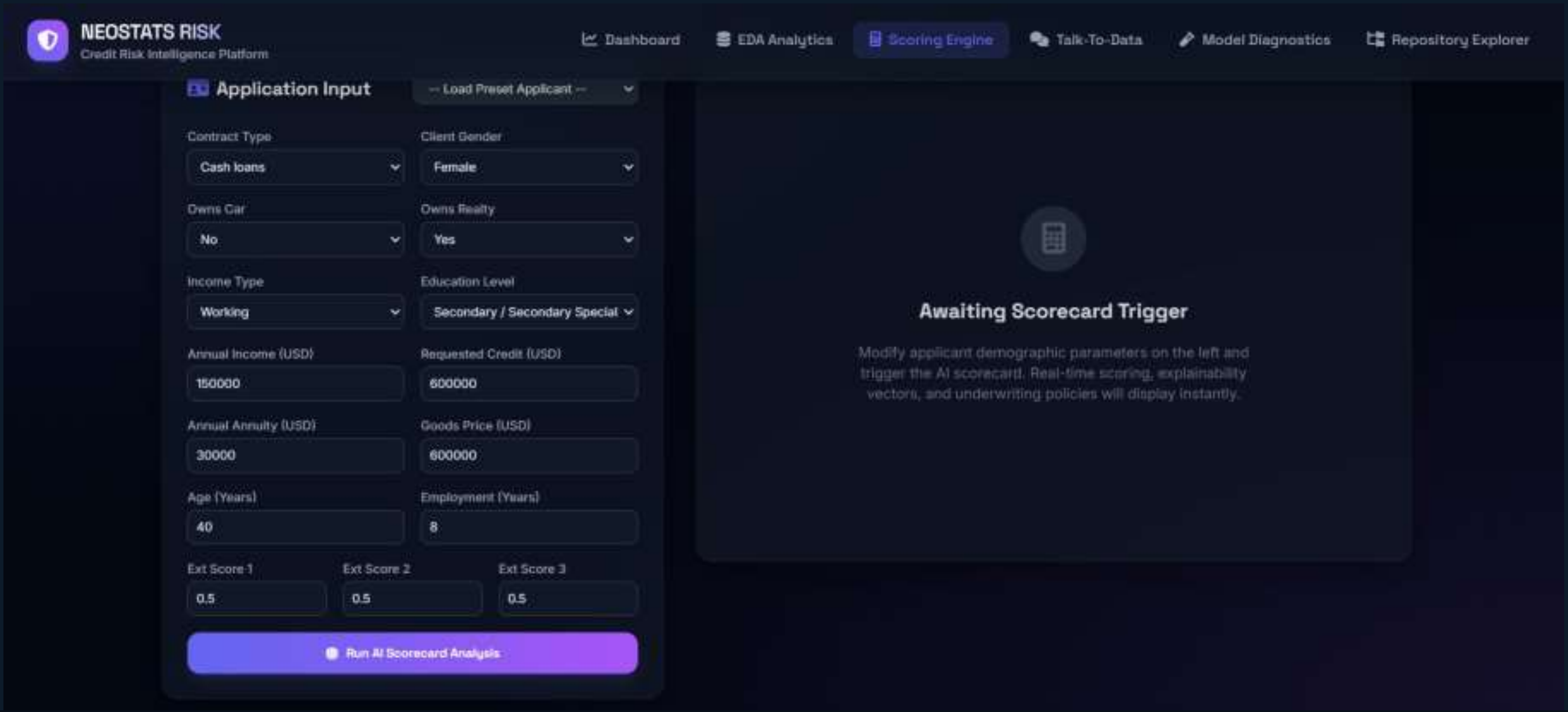
EDA Analytics — Portfolio Deep Dive

Live App Screen

Default rate by education, income, occupation · Income binning margin separation chart

Scoring Engine — Real-Time Risk Assessment

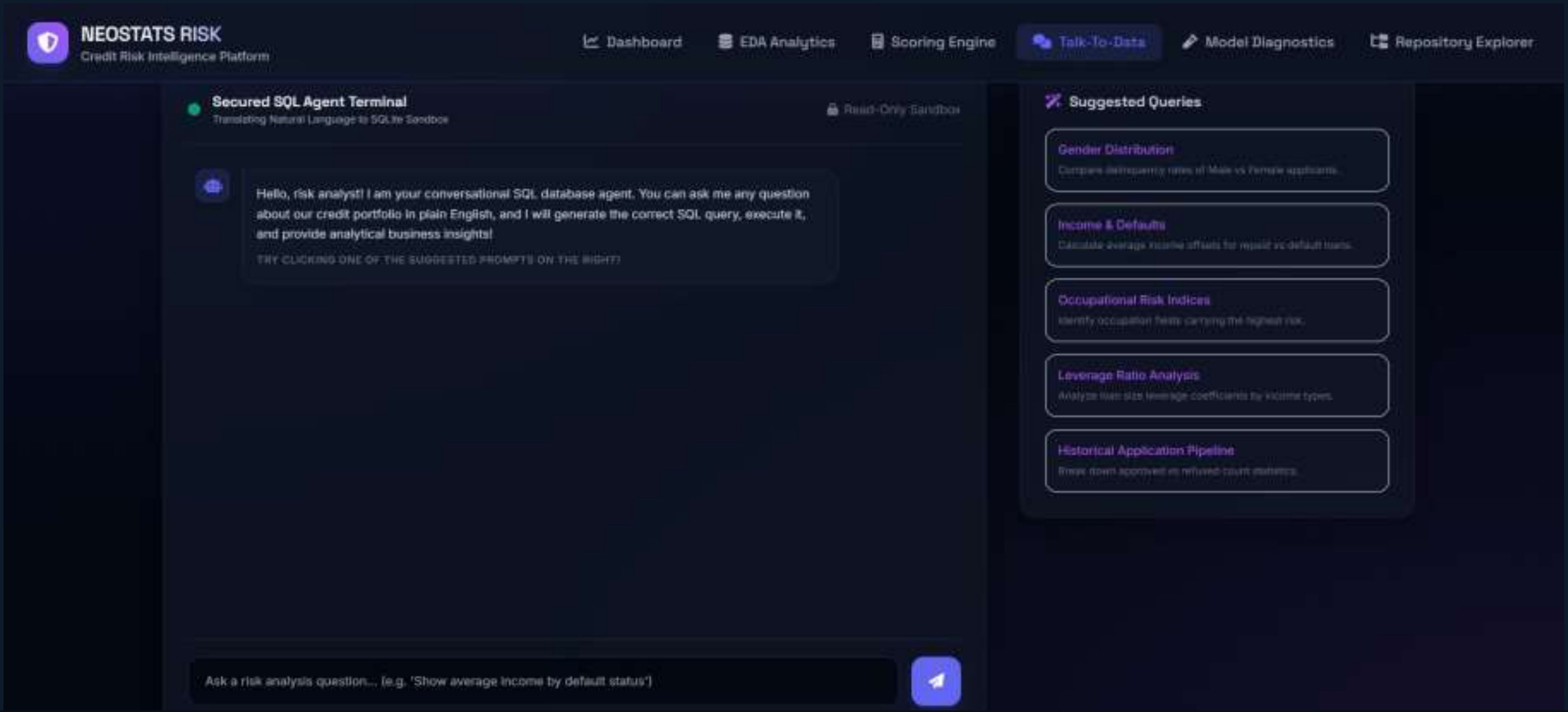
Live App Screen



Input applicant details → Run AI Scorecard Analysis → Get risk band (Low / Medium / High) instantly

Talk-to-Data — NL-to-SQL Chatbot

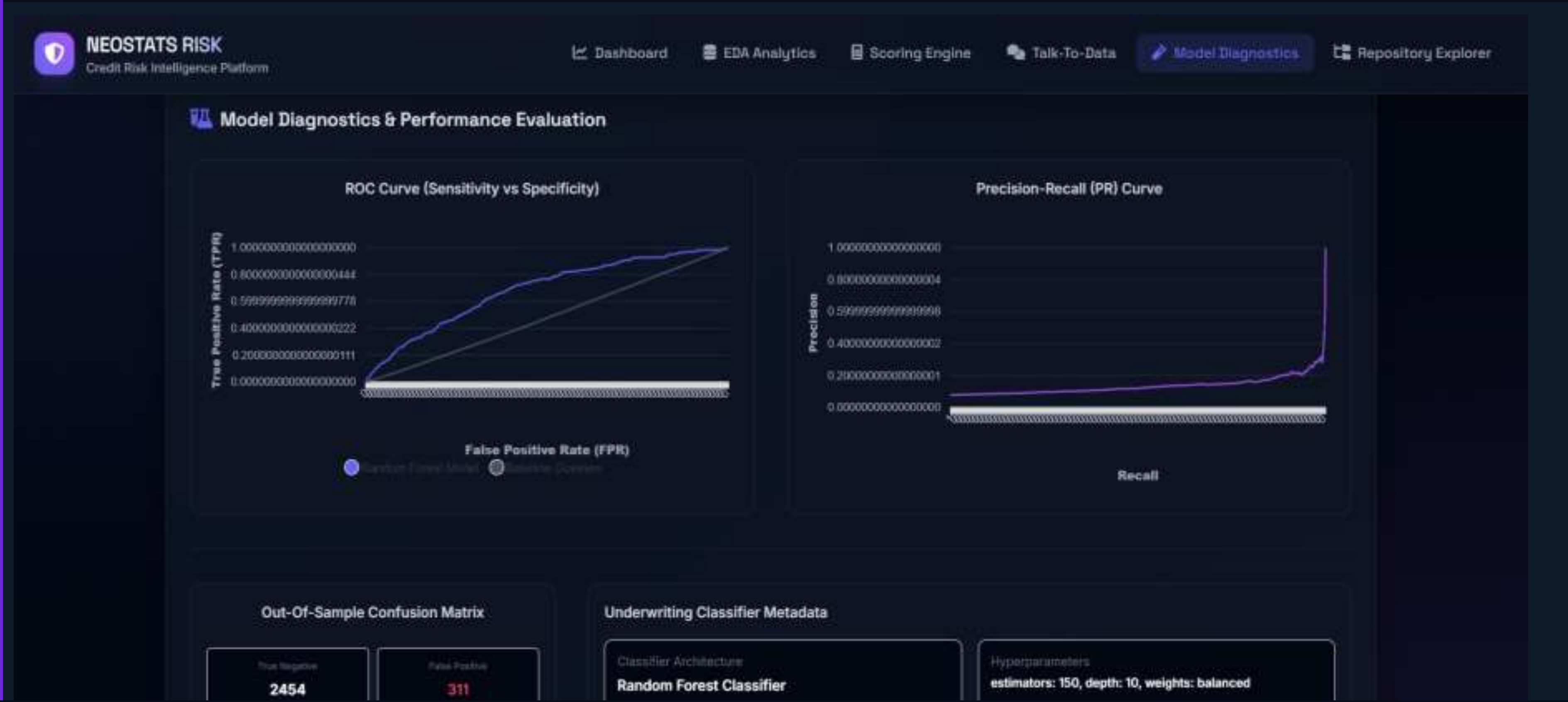
Live App Screen



5 pre-built queries: Gender distribution · Income & defaults · Occupational risk · Leverage ratio · Historical pipeline

Model Diagnostics & Performance Evaluation

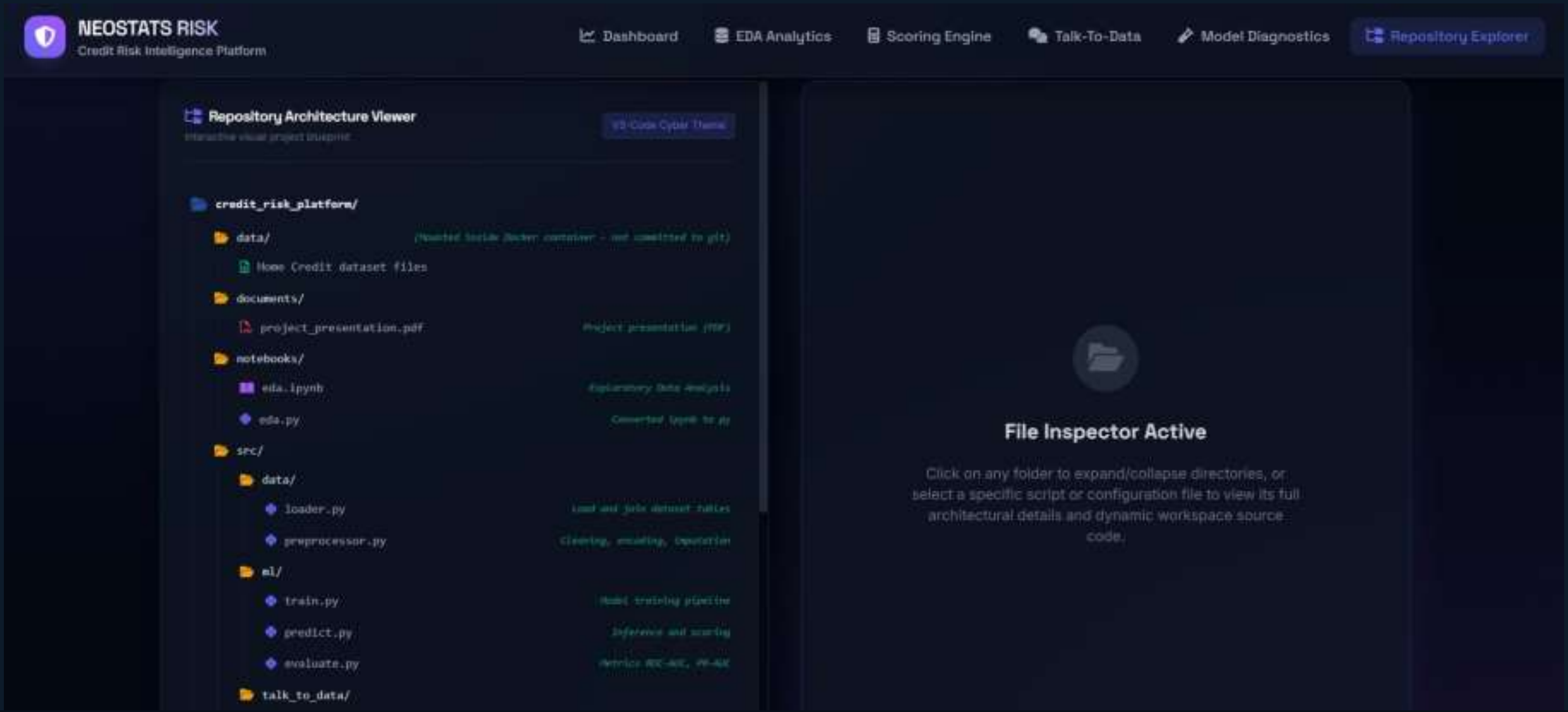
Live App Screen



ROC Curve · Precision-Recall Curve · Confusion Matrix · Random Forest (150 estimators, depth 10, balanced weights)

Repository Explorer — Architecture Viewer

Live App Screen



Interactive VS-Code style project blueprint · Click any file to view its architecture details and source

Machine Learning Layer — How It Works

Algorithm	Random Forest Classifier (150 estimators, depth 10)
Class Imbalance	class_weight: balanced — ensures minority class not ignored
Feature Engineering	120+ features: demographics, bureau data, credit history
Validation	Out-of-sample confusion matrix · ROC-AUC · PR-AUC
Output	Risk Score 0–100 → Low / Medium / High band

LOW · Score 0–35 **Approve**	**MEDIUM · Score 36–65** **Manual Review**	**● HIGH · Score 66–100** **Decline**

Major Design Decisions

01 Random Forest over Deep Learning

Interpretability and speed matter more than marginal accuracy gains. Random Forest produces reliable probability outputs and is directly compatible with SHAP — making it audit-ready and production-safe.

02 SQLite for Development, PostgreSQL-ready

SQLite keeps setup friction near zero — the evaluator doesn't need a database server. The data layer is abstracted so swapping to PostgreSQL in production requires only a config change.

03 LLM-Powered NL-to-SQL with Guard Rails

Natural language queries are converted via LLM prompts with strict READ-ONLY sandbox enforcement. Prompt templates are versioned to prevent hallucinations and SQL injection.

04 class_weight: balanced — No SMOTE Overhead

Oversampling (SMOTE) adds training complexity and can introduce noise. Balanced class weights handle the 8% default imbalance cleanly without synthetic samples, keeping the pipeline lean.

Architecture & Tech Stack

UI Layer · Multi-tab web dashboard (Dashboard · EDA · Scoring · Talk-to-Data · Diagnostics · Repo Explorer)

Application Layer · FastAPI backend · LLM NL-to-SQL agent · SHAP explainer · Decision rules engine

ML Layer · Random Forest Classifier · Feature engineering pipeline · Model training & inference

Data Layer · SQLite (dev) / PostgreSQL (prod) · Home Credit CSV · Preprocessing & imputation

ML

Random Forest · Scikit-learn

XAI

SHAP · LIME

LLM

Claude / OpenAI API

Backend

Python · FastAPI

Frontend

Streamlit / React

Deploy

Docker · Compose

```
$ git clone <repo>  &&  docker-compose up  →  Full platform running in minutes
```

What We Built

- 15,000 applications analyzed · 7.83% default rate identified
- Random Forest with 120+ features · balanced class weights
- SHAP explainability on every prediction — audit-ready
- NL-to-SQL chatbot · 5+ verified working queries
- Dockerized · runs with a single command
- Clean modular code · well-documented README

Intelligence.
Innovation.
Impact.

Thank You